# Email Threat Trends: How Attackers Are Reinventing Email Attacks

Meta: Gain insights into email threat trends with calculations that adapt to yearly sample size changes, ensuring reliable analysis.

Slug: email-threat-trends

Email threat trends are moving faster than most teams' playbooks. Email has always been the quiet workhorse of business communication, but it has now become the frontline for almost every serious email security threat, from ransomware entry points to credential theft and data exfiltration.

Attackers aren't wasting time on noisy spam or clumsy hoaxes anymore. They combine AI-written lures, subtle **email threat scams**, and weaponized "harmless" files to build attacks that look like everyday business traffic, right up until someone clicks.

What our Cybersecurity Report 2026 data makes clear is that these email threat trends are evolving faster than traditional controls such as static filters or once-a-year training. Ransomware is resurging, multi-vector campaigns are spreading across identities and SaaS platforms, and gaps in email threat detection and response are being exploited at scale.

In the next sections, we'll unpack the trends behind the numbers, show how new techniques, automation, and credential compromise interact, and outline what modern email threat defense should look like.

## New Email Threat Trends Shaping 2025

Email remains the backbone of business communication and, as our data shows, it also continues to be the primary battleground for attackers. 2025's classification and threat-type shifts reveal two simultaneous realities: attackers are experimenting with new filetypes and low-effort delivery methods (TXT and legacy DOC surged), and at the same time social engineering remains a consistent lever for compromise.

Put simply: quantity and quality are changing. While classic spam volumes have stabilized after normalization, higher-impact categories (Malware, Scam, Phishing, etc.) are growing substantially. That combination (more dangerous content delivered at scale) increases the likelihood that even well-defended organizations will face incidents unless they adjust detection, user awareness, and recovery practices.

## Spam, Malware, & Advanced Threat Metrics

The headline numbers are unambiguous: **Malware** saw the largest relative increase (+130.92%), followed by **Scams** (+34.70%) and **Phishing** (+20.97%). Those three categories account for the bulk of the risk that results in operational impact (data theft, encryption, business disruption).

Meanwhile, categories that traditionally represented lower business risk; legitimate Messages, Transactional, and Commercial Email moved only modestly, indicating that malicious actors are concentrating effort on higher-value attack types.

Key implications:

- **Proliferation of malicious payloads.** A 131% jump in Malware classification means more emails are carrying active payloads (or at least payload indicators) rather than simple noise. Detection strategies must assume malicious intent either way.
- **Scams and advanced social engineering are on the rise.** Scams (+34.7%) coupled with Phishing (+21.0%) signals that attackers are refining their lures and ROI. They're making more convincing frauds, and more customized messages, likely enabled by generative AI technologies.
- **"Dirty Commercial" growth undermines heuristic filters.** Dirty Commercial Emails (+17.72%) suggests attackers may be weaponizing lower-quality marketing templates to evade simple content filters and blend in with legitimate marketing traffic.
- **Targeted spear-phishing share is down, but not gone.** Suspect / Spear-Phishing is down (-9.75%), which likely reflects a shift to more automated / commodity phishing *and* to credential-theft approaches that bypass classic spear phishing detection. Don't be lulled into complacency: targeted attacks remain high-impact even at lower volume.

### Email Classification Categories

| Category | Adjusted YoY Change 2025 vs. 2024 |
|---|---|
| Malware | +130.92% |
| Scam | +34.70% |
| Phishing | +20.97% |
| Dirty Commercial Emails | +17.72% |
| Commercial Email | +2.37% |

| | |
|---|---|
| Legitimate Messages | +3.38% |
| Transactional | +3.19% |
| Spam | +0.03% |
| Social | -8.05% |
| Suspect / Spear Phishing | -9.75% |
| Pro Commercial Emails | -13.73% |
| Bounce | -18.69% |

**NOTE:** Calculations take into account and adjust for sample size changes from year to year.

## Category Classification Descriptions

### *Spam*

Unsolicited bulk email messages sent to a large number of recipients, typically for advertising or malicious purposes.

### *Phishing*

Fraudulent emails designed to trick recipients into revealing sensitive information such as passwords, credit card numbers, or personal data.

### *Commercial Email*

Legitimate marketing or promotional emails sent by businesses to customers or prospects, often for product announcements or offers.

### *Legitimate Messages*

Authentic, non-promotional emails exchanged between individuals or organizations for normal communication purposes.

### *Pro Commercial Emails*

Professional-grade marketing emails, often highly targeted and personalized, typically used in B2B campaigns.

### *Transactional*

Emails triggered by user actions or system events, such as order confirmations, password resets, or account notifications.

### *Social*

Emails originating from social media platforms, including notifications, friend requests, and activity alerts.

### *Bounce*

Emails that fail to deliver to the recipient's inbox due to invalid addresses, full mailboxes, or server issues.

### *Dirty Commercial Emails*

Marketing emails that violate compliance standards or best practices, often poorly formatted or misleading.

### *Scam*

Emails intended to defraud recipients, often involving fake offers, lottery winnings, or impersonation schemes.

### *Malware*

Emails containing malicious attachments or links designed to install harmful software on the recipient's device.

### *Suspect / Spear Phishing*

Highly targeted phishing attempts aimed at specific individuals or organizations, often using personalized details to appear credible.

## Attack Techniques Used in Email Attacks 2025

The 2025 attack-technique landscape shows a clear preference for **evasion-first** tactics: attackers are less focused on single flashy payloads and more on slipping past filters and human suspicion.

The top techniques: header forgery, subtle HTML tricks, use of legitimate hosting, and URL obfuscation are all optimized to blend malicious intent into otherwise benign-looking mail.

That shift explains why we're seeing fewer obvious spear-phish samples but more successful credential-theft and multi-stage intrusions: the email is the first step, not the punchline.

Key observations:

- **Header and metadata manipulation dominate.** Fake `From` and manipulated spam-related headers top the list, demonstrating that spoofing and metadata tampering remain low-cost, high-impact methods to defeat naive filtering and trigger human trust.
- **Abuse of legitimate infrastructure is rising.** Sending campaigns via reputable hosting platforms makes malicious mail appear to come from trustworthy sources. This is a tactic that increases deliverability and reduces immediate filter suspicion.
- **URL obfuscation is ubiquitous.** URL shortening, non-ASCII characters, exotic TLDs (Top Level Domains), and domain fuzzing are all simple ways to hide destination intent and bypass blocklists or visual inspection.
- **HTML / MIME tricks aim to confuse detectors, not readers.** Empty `<a>` tags, multi-part messages, and zero-(size)-font insertion are designed to mislead signature and keyword-based scanning engines while preserving readability for recipients.
- **Automated, high-volume evasion beats small-scale targeting.** These techniques scale: attackers can roll out many campaigns that individually look benign but collectively yield credential captures, account compromise, or chained downloads.

## Top 10 Attack Techniques Used in Email Attacks in 2025

| Rank | Technique |
|---|---|
| 1 | Fake From Header Alteration |
| 2 | Fake Spamcause Header Alteration |
| 3 | Leverage Legit Hosting Platform to Send Campaign |
| 4 | Use of Exotic or Non-Existent TLDs |
| 5 | URL Shortening |
| 6 | HTML `<a>` Tag Empty |
| 7 | Multi-Parted Emails |
| 8 | URL with Non-ASCII Characters |
| 9 | Random Domains / URL Fuzzing |
| 10 | ZeroFont Technique |

Technique Descriptions

## 1. Fake From Header Alteration

Attackers forge the “From” header in emails to impersonate trusted senders, tricking recipients into believing the email is legitimate.

## 2. Fake Spamcause Header Alteration

Manipulation of spam-related headers to bypass spam filters and make malicious emails appear safe.

## 3. Leverage Legit Hosting Platform to Send Campaign

Using reputable hosting or email services (e.g., cloud platforms) to distribute phishing or malicious campaigns, making detection harder.

## 4. Use of Exotic or Non-Existent TLDs

Employing unusual or fake top-level domains (e.g., `.xyz`, `.club`) to create deceptive URLs that look legitimate.

## 5. URL Shortening

Using URL shorteners (e.g., bit.ly) to hide the true destination of malicious links, making them harder to detect.

## 6. HTML <a> Tag Empty

Embedding empty anchor tags in HTML emails to confuse spam filters or hide malicious links.

## 7. Multi-Parted Emails

Sending emails with multiple MIME parts (e.g., text and HTML) to evade detection by security tools.

## 8. URL with Non-ASCII Characters

Including special or Unicode characters in URLs to create visually deceptive links (e.g., homoglyph attacks).

### 9. Random Domains / URL Fuzzing

Generating random or slightly altered domains to bypass domain-based filtering and detection systems.

### 10. ZeroFont Technique

Inserting zero-size font text in emails to manipulate keyword-based filters while keeping the message readable to humans.

## Attachment Use and Types in Attacks

Attachment trends in 2025 demonstrate a pronounced pivot in malware delivery strategy. The fastest-growing file carriers are **TXT (+181.39%)** and **DOC (+118.25%)**, with ZIP and modern Office formats (DOCX, XLSX) also present but growing more modestly.

Legacy or once-popular vectors (HTML, RAR, HTM, XLS) declined, while **ICS** and **SHTML** appear as new entries to our top-ten list. This is a sign attackers are searching for overlooked or under-inspected file types plus calendar files or server-side include vectors.

Key takeaways:

- **TXT and legacy DOC are alarm bells.** TXT files, which are widely treated as “low risk”, are being weaponized as staging artifacts (containing obfuscated URLs or scripts). Legacy DOCs (with macro support) remain attractive because many environments still allow or fail to inspect office macros aggressively.
- **Archives STILL matter.** ZIP (+29.82%) remains a vehicle for payload bundling and evasion; compressed archives continue to be a reliable attacker tactic.
- **Emergence of ICS and SHTML is noteworthy.** Calendar invites (ICS) and server-include variants (SHTML) represent non-traditional vectors that can bypass some mail filters and user expectations. This is especially true for recipients who accept calendar items or preview HTML content.
- **Decline in HTML/HTM/RAR/XLS likely reflects defensive hardening**, but attackers are redirecting to less-monitored channels rather than abandoning email as a vector.

## File-Types for Malicious Payloads 2025

| File Type | Adjusted YoY Change 2025 vs. 2024 |
|---|---|

| | |
|---|---|
| TXT | +181.39% |
| DOC | +118.25% |
| ZIP | +29.82% |
| DOCX | +11.69% |
| XLSX | +7.85% |
| PDF | -3.32% |
| HTML | -27.44% |
| RAR | -36.93% |
| HTM | Dropped from top 10 |
| XLS | Dropped from top 10 |
| ICS | New Entry to list in 2025 |
| SHTML | New Entry to list in 2025 |

**NOTE:** Calculations take into account and adjust for sample size changes from year to year.

File Type Definitions

## PDF

**Portable Document Format** – Commonly used for documents; attackers often embed malicious links or scripts within PDFs.

## DOC

**Microsoft Word Document (Legacy)** – Older Word file format; can contain macros that execute harmful code.

## DOCX

**Microsoft Word Document (Modern)** – Current Word format; supports embedded macros and scripts that can be exploited.

## XLS

**Microsoft Excel Spreadsheet (Legacy)** – Older Excel format; often targeted for macro-based attacks.

## XLSX

**Microsoft Excel Spreadsheet (Modern)** – Current Excel format; can include malicious macros or links.

## TXT

**Plain Text File** – Simple text files; attackers may use them to deliver phishing content or scripts disguised as text.

## HTML

**HyperText Markup Language File** – Web page format; often used in phishing emails with embedded malicious links.

## HTM

**HyperText Markup Language File (Variant)** – The legacy file extension for HTML files; used for web content and phishing payloads.

## SHTML

**Secure HTML File** – HTML variant supporting server-side includes; can be exploited for malicious redirects.

## ZIP

**Compressed Archive File** – Commonly used to bundle files; attackers hide malware inside compressed archives.

## RAR

**Compressed Archive File (Alternative)** – Similar to ZIP but uses different compression algorithm; often used for malware delivery.

## ICS

**Calendar File** – iCalendar format; attackers use malicious calendar invites to deliver phishing links or payloads.

## Protect Your Microsoft 365 Environment from Today's Evolving Email Threats

Ransomware's resurgence and shifting email threat trends show that native controls in Microsoft 365 aren't enough. You need dedicated email threat protection that can keep up with evasive tactics, identity abuse, and multi-vector campaigns.

Hornetsecurity's 365 Total Protection adds advanced threat protection for email on top of Microsoft 365, combining AI-driven filtering and sandboxing to stop suspicious messages early.

It includes an AI-driven email threat scanner that inspects

- fake headers,
- weaponized TXT files,
- and obfuscated URLs

before users ever see them.

Backed by global email threat intelligence, the service continuously learns from new campaigns so your policies and detections stay ahead of attackers.

With automated email threat protection you can keep modern attacks out of user inboxes and safeguard your Microsoft 365 environment. Discover 365 Total Protection today.

## Conclusion: Turning Email Threat Trends into Action

Taken together, this year's data paints a simple but uncomfortable picture: email attacks are getting quieter, smarter, and better at slipping past legacy controls. The most dangerous email threat trends are the ones that hide inside normal-looking conversations, file shares, and workflows rather than shouting for attention.

From forged headers and abused hosting platforms to weaponized TXT files and calendar invites, attackers are optimizing every step of the delivery chain to evade filters and exploit trust. Organizations that still rely on aging gateways or minimal training are effectively giving advanced actors more time to experiment and refine their playbooks.

Closing that gap means treating email as one part of a broader ransomware and identity ecosystem: layered controls, strong authentication, continuous monitoring, and user

awareness that reflects today's reality, not the email fraud threat patterns from years ago.

Email isn't going away, and neither are attackers, but you're not powerless. With the right mix of policy, education, and modern email threat protection platforms, security teams can turn raw data about email threat trends into practical defenses that keep business running—even as attackers evolve.

## FAQ: Email Threat Trends and Modern Ransomware

### Why are TXT and legacy DOC files suddenly so prominent in recent email threat trends?

Because attackers are shifting toward file types that many controls consider low risk. TXT files often slip through as plain text but can hide obfuscated URLs or scripts that kick off a multi-stage email security threat, while legacy DOC files still support macros in many environments and are ideal carriers for stealthy payloads.

### What makes these new email attacks and email threat trends harder to detect?

Modern campaigns are built around evasion rather than obvious payloads. Threat actors blend forged headers, legitimate hosting, multi-part messages, exotic TLDs, ZeroFont, and other tricks to confuse filters and analysis tools, so the email itself looks benign while quietly enabling credential theft, ransomware deployment, or later-stage compromise.

### If spear-phishing appears to be decreasing in the data, does that mean targeted email attacks are fading?

Not at all. The dip in classic spear-phishing simply reflects a pivot toward automated credential theft and large-scale phishing powered by better tooling and AI. High-value, human-crafted attacks are still very much in play—they just sit alongside a much larger volume of scalable campaigns, which is exactly why organizations need continuous visibility into email threat trends and strong controls to catch the handful that really matter.